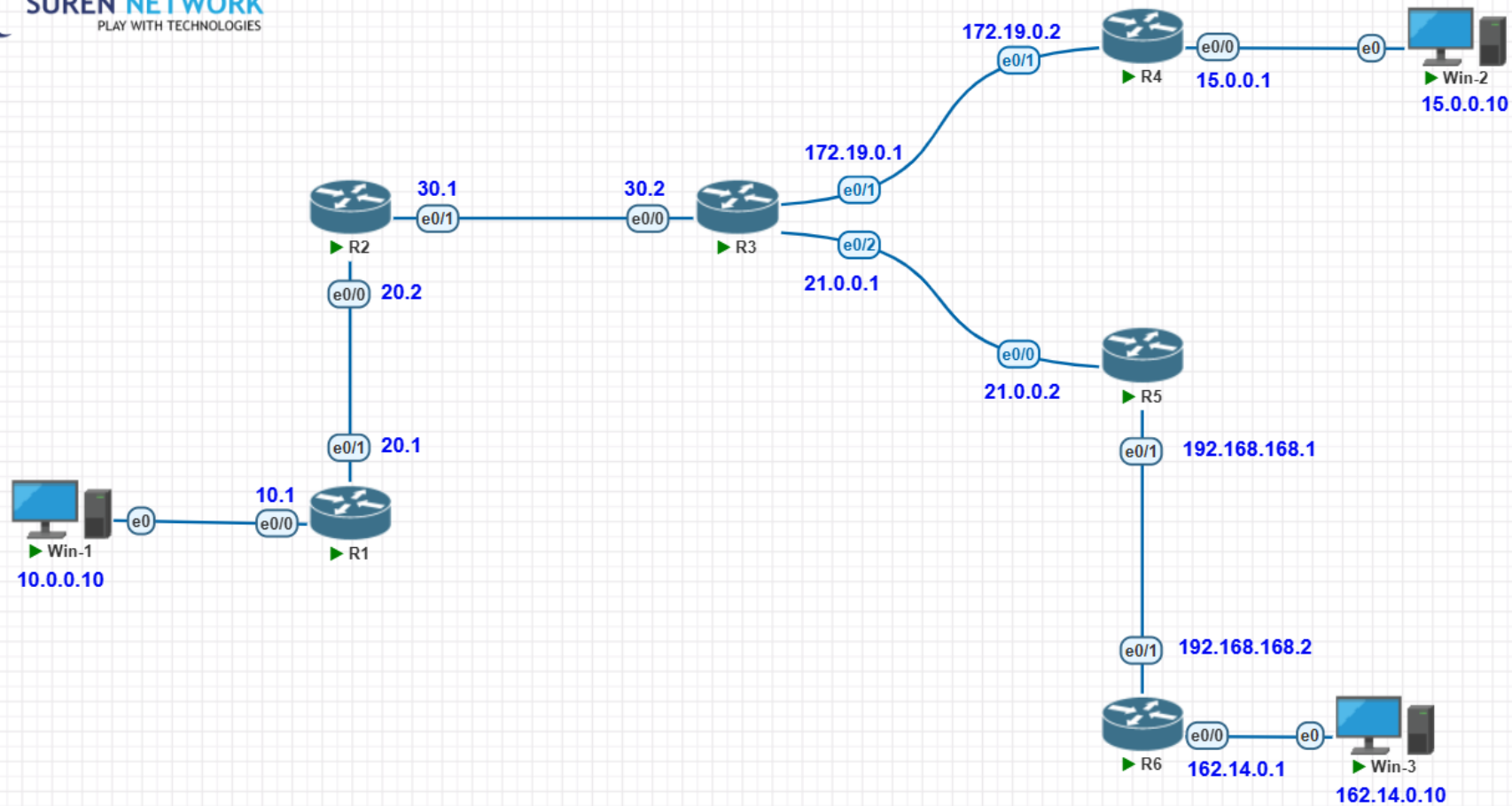




+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.0.0.0/8 is directly connected, Ethernet0/0

L 10.0.0.1/32 is directly connected, Ethernet0/0

S 15.0.0.0/8 [1/0] via 20.0.0.2

20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 20.0.0.0/8 is directly connected, Ethernet0/1

L 20.0.0.1/32 is directly connected, Ethernet0/1

S 21.0.0.0/8 [1/0] via 20.0.0.2

S 30.0.0.0/8 [1/0] via 20.0.0.2

S 162.14.0.0/16 [1/0] via 20.0.0.2

S 172.19.0.0/16 [1/0] via 20.0.0.2

S 192.168.168.0/24 [1/0] via 20.0.0.2

R1(config)#do sh ip route static

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S 15.0.0.0/8 [1/0] via 20.0.0.2

S 21.0.0.0/8 [1/0] via 20.0.0.2

S 30.0.0.0/8 [1/0] via 20.0.0.2

S 162.14.0.0/16 [1/0] via 20.0.0.2

S 172.19.0.0/16 [1/0] via 20.0.0.2

S 192.168.168.0/24 [1/0] via 20.0.0.2

R1(config)#

Manually added a static route to the router (R1)

Manually added a static route to the router (R6)

labs.surennetwork.com/store/public/admin/labs/terminal?terminals=eyJ0YWZlZjpbeyJuYW1lIjoV2luLTEiLCJ1c...

linkedin | Webmail - Main | Networking Acade... | Online typing WPM... | NCCR Biometric Att... | HP Product Warrant... | Fujitsu Technical Su... | D-LINK | HP » Device Status | All Bookmarks

Win-1 x R1 x R2 x R3 x

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 30.0.0.1
S 15.0.0.0/8 [1/0] via 172.19.0.2
S 20.0.0.0/8 [1/0] via 30.0.0.1
21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 21.0.0.0/8 is directly connected, Ethernet0/2
L 21.0.0.1/32 is directly connected, Ethernet0/2
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/0
L 30.0.0.2/32 is directly connected, Ethernet0/0
S 162.14.0.0/16 [1/0] via 21.0.0.2
172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.19.0.0/16 is directly connected, Ethernet0/1
L 172.19.0.1/32 is directly connected, Ethernet0/1
S 192.168.168.0/24 [1/0] via 21.0.0.2

R3(config)#do sh ip route static
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 30.0.0.1
S 15.0.0.0/8 [1/0] via 172.19.0.2
S 20.0.0.0/8 [1/0] via 30.0.0.1
S 162.14.0.0/16 [1/0] via 21.0.0.2
S 192.168.168.0/24 [1/0] via 21.0.0.2

R3(config)#

Manually added a static route to the router (R3)

35°C
Partly sunny

Search

03:29 pm
02-09-2026

labs.surennetwork.com/store/public/admin/labs/terminal?terminals=eyJ0YWJzljpbeyJuYW1lIjoIV2luLTEiLCJ1c...

Win-1 x R1 x R2 x R3 x R4 x

```
R4(config)#do sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              15.0.0.1        YES manual  up          up
Ethernet0/1              172.19.0.2      YES manual  up          up
Ethernet0/2              unassigned      YES TFTP   down        down
Ethernet0/3              unassigned      YES TFTP   down        down

R4(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 172.19.0.1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    15.0.0.0/8 is directly connected, Ethernet0/0
L    15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
L    172.19.0.2/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 172.19.0.1
R4(config)#do sh ip route
```

Manually added a static route to the router (R4)

35°C Partly sunny 03:29 pm 02-09-2026

The screenshot shows a terminal window with the following content:

```
R5(config)#do sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              21.0.0.2        YES manual  up          up
Ethernet0/1              192.168.168.1   YES manual  up          up
Ethernet0/2              unassigned      YES TFTP   down        down
Ethernet0/3              unassigned      YES TFTP   down        down

R5(config)#do sh ip route static
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 21.0.0.1
S    15.0.0.0/8 [1/0] via 21.0.0.1
S    20.0.0.0/8 [1/0] via 21.0.0.1
S    30.0.0.0/8 [1/0] via 21.0.0.1
S    162.14.0.0/16 [1/0] via 192.168.168.2
S    172.19.0.0/16 [1/0] via 21.0.0.1
R5(config)#
```

A large orange bracket is drawn on the right side of the terminal output, grouping the static routes. A text annotation "Manually added a static route to the router (R5)" is placed to the right of the bracket, pointing to the static routes.


```
R6(config)#do sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              162.14.0.1      YES manual  up          up
Ethernet0/1              192.168.168.2   YES manual  up          up
Ethernet0/2              unassigned      YES TFTP   down        down
Ethernet0/3              unassigned      YES TFTP   down        down

R6(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 192.168.168.1
S    15.0.0.0/8 [1/0] via 192.168.168.1
S    20.0.0.0/8 [1/0] via 192.168.168.1
S    21.0.0.0/8 [1/0] via 192.168.168.1
S    30.0.0.0/8 [1/0] via 192.168.168.1
     162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C     162.14.0.0/16 is directly connected, Ethernet0/0
L     162.14.0.1/32 is directly connected, Ethernet0/0
S    172.19.0.0/16 [1/0] via 192.168.168.1
     192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.168.0/24 is directly connected, Ethernet0/1
L     192.168.168.2/32 is directly connected, Ethernet0/1
R6(config)#do sh ip route static
```

Manually added a static route to the router (R6)

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>cd \

C:\>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 10.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1

C:\>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time=1ms TTL=255
Reply from 10.0.0.1: bytes=32 time=1ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\>
```

Successfully reached all the PC's IPs


```
C:\WINDOWS\system32\cmd.exe
Subnet Mask . . . . . : 255.0.0.0
Default Gateway . . . . . : 15.0.0.1

C:\>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=1ms TTL=124

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 15.0.0.1

Pinging 15.0.0.1 with 32 bytes of data:

Reply from 15.0.0.1: bytes=32 time<1ms TTL=255
Reply from 15.0.0.1: bytes=32 time<1ms TTL=255
Reply from 15.0.0.1: bytes=32 time<1ms TTL=255
Reply from 15.0.0.1: bytes=32 time=1ms TTL=255

Ping statistics for 15.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=1ms TTL=124
Reply from 162.14.0.10: bytes=32 time=5ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 5ms, Average = 2ms

C:\>ping 20.0.0.1

Pinging 20.0.0.1 with 32 bytes of data:

Reply from 20.0.0.1: bytes=32 time=1ms TTL=252
Reply from 20.0.0.1: bytes=32 time=1ms TTL=252
Reply from 20.0.0.1: bytes=32 time=1ms TTL=252
Reply from 20.0.0.1: bytes=32 time=1ms TTL=252

Ping statistics for 20.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

Successfully reached all the PC's IPs

labs.surennetwork.com/store/public/admin/labs/terminal?terminals=eyJ0YWJzljpbeyJuYW1lIjoV2luLTEiLCJ1c...

Win-1 x R1 x R2 x R3 x R4 x R5 x R6 x Win-2 x Win-3 x

C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>cd \

C:\>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=2ms TTL=123

Reply from 10.0.0.10: bytes=32 time=1ms TTL=123

Reply from 10.0.0.10: bytes=32 time=2ms TTL=123

Reply from 10.0.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 10.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=1ms TTL=124

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 20.0.0.1

Pinging 20.0.0.1 with 32 bytes of data:

Reply from 20.0.0.1: bytes=32 time=2ms TTL=251

Reply from 20.0.0.1: bytes=32 time=2ms TTL=251

Reply from 20.0.0.1: bytes=32 time=2ms TTL=251

Reply from 20.0.0.1: bytes=32 time=4ms TTL=251

Ping statistics for 20.0.0.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time<1ms TTL=128

Reply from 162.14.0.10: bytes=32 time<1ms TTL=128

Reply from 162.14.0.10: bytes=32 time<1ms TTL=128

Reply from 162.14.0.10: bytes=32 time<1ms TTL=128

Ping statistics for 162.14.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

Successfully reached all the PC's IPs

Start

C:\WINDOWS\system...

11:04 AM

35°C

Partly sunny

03:34 pm

02-09-2026